

## Classification and Decision Trees Activity

### 1. Tree Structures for Classification



A tree structure can be created to perform classification. For example, the tree on the left is used to predict whether someone who borrows money from a bank will default on the loan.

Use the tree to classify a arrived examples with the following features:

mini *MarSt*  
NO

| Homeowner | Marital Status | Income | Loan Will Default? |
|-----------|----------------|--------|--------------------|
| no        | married        | 80K    | ??                 |
| no        | divorced       | 93K    | ??                 |

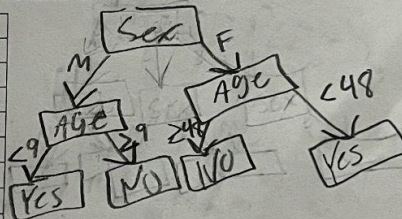
*MarSt* *Single*  
*Home Owner* *NO*

Trace the pathway through the tree for these examples. What labels (yes or no) does the tree assign?

### 2. Decision Tree Construction

The following is a subset of the Titanic Survivors dataset available from Kaggle <https://www.kaggle.com/c/titanic/>. Draw a decision tree that correctly classifies each item.

| Pclass | Sex    | Age | Survived |
|--------|--------|-----|----------|
| 3      | male   | 14  | No       |
| 2      | female | 25  | Yes      |
| 2      | male   | 35  | No       |
| 3      | female | 24  | Yes      |
| 3      | female | 48  | No       |
| 2      | female | 45  | Yes      |
| 2      | female | 22  | Yes      |
| 1      | male   | 62  | No       |
| 2      | male   | 3   | Yes      |
| 3      | male   | 9   | No       |
| 2      | male   | 2   | Yes      |
| 2      | female | 36  | Yes      |



*Agcs:*  
2. Y  
2. Y  
9. Y  
11. N  
22. Y  
24. Y  
25. Y  
35. Y  
36. Y  
45. Y  
48. Y  
62. N

Now use your tree to classify the items in this test set:

| Pclass | Sex    | Age | Survived |
|--------|--------|-----|----------|
| 1      | female | 17  | YES      |
| 3      | female | 41  | YES      |
| 3      | male   | 17  | NO       |
| 3      | male   | 1   | YES      |

### 3. Tree Reflection

In this case, it was straightforward to develop a decision tree that could perfectly classify the training set by "eyeballing" the data. Consider the case where we have thousands of elements in our training set, each with dozens of attributes.

- Given that there are potentially many possible trees that fit the training data, how might you go about deciding between them? What constitutes a "good" tree?

A good tree would have a larger dataset and use probability to decide what the probability of each leaf node being the target is given the preceding features.

- How could you automate the process of tree construction?

by writing an algorithm to view the data points and conclude which features offer more concrete conclusions about the data.